Platoon Control of Autonomous Vehicles

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***Abstract*—Autonomous Vehicle Platoons have been shown to improve traffic density, fuel efficiency, and road safety compared to human-driven vehicles. This report considers the control of multi-lane platoons to solve the lane-changing problem in a coordinated fashion while maintaining string stability and platoon tightness. For distance control, we propose a novel controller, a modified version of the traditional CACC controller: the relative distance is always maintained to equal a calculated "Safe Distance". A second PID-based controller is also implemented to drive the orientation error to zero as the lane change manoeuvre finishes.**

***Index Terms* — Autonomous Vehicles, Multi-Agent Systems, Platoon Control**

I. Introduction

THE aim of platoon control is to maintain a safe distance between Connected Autonomous Vehicles and also ensure coordinated lane change of the whole platoon. Similar research is presented in [1] where the authors propose a FSM based decision-making process for Lane Change Maneuver. [2] discusses various existing control and analysis approaches for string stability of 1-D platoons. Various communication topologies can be explored for better control performance [3]. We use Predecessor-Following V2V model in this work. Cooperative Adaptive Cruise Control (based on Model Predictive Control), as proposed in [4], has been employed for longitudinal control and a PID controller for vehicle alignment and the model has been simulated in MATLAB.

II. Methodology and Technical Contribution

A. Coordinated Adaptive Cruise Control

Cruise control is present in most commercially available vehicles. The developed algorithm uses this feature, thus eliminating the need for additional infrastructure. The objective is to maintain a velocity-dependent safe distance ($d_{safe} = d_{default} + T_{gap}V_{ego}\cos\lambda_{ego}$) between 2 vehicles, where λ_{ego} can be seen from fig. 1. We use Model Predictive Control for implementing an adaptive cruise control technique for a single vehicle [4] and extend this controller to a platoon of CAVs. As depicted in [4], the follower or ego vehicle is programmed for two behaviors switched according to its distance from the preceding vehicle (d_{rel}). However, instead of tracking a constant set velocity, we have programmed the ACC model for a variable input V_{set} , to track d_{safe}.

$$V_{set} = V_{lead} + K_p e_d + K_d \frac{de_d}{dt} + K_I \int_0^t e_d dt$$

Where $e_d = d_{rel} - d_{safe}$. Hence, the proposed controller always tries to maintain $d_{rel} = d_{safe}$.

B. Coordinated Lane Change Manoeuvre

Lane changing involves additional challenges to the longitudinal control, since the orientation of the vehicles also needs to be controlled along with the relative distance between them. For distance control, all the velocities and distances between adjacent vehicles have been computed along the line joining them. As a result, the scenario can be represented using a 1-D control problem, similar to the last section, even if the vehicles have different orientations.

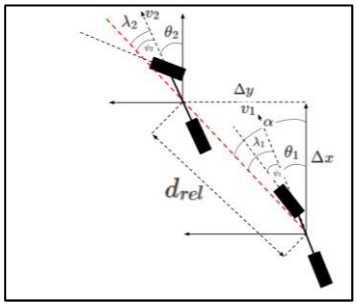


Fig 1. Vehicle poses during lane change

For orientation control during the lane change, we use a PID controller that calculates the required steering angle using the difference in orientations of the adjacent vehicles as the error signal.

$$\psi_1 = K\psi_2 + K_p e_\theta + K_d \frac{de_\theta}{dt} + K_I \int_0^t e_\theta dt$$

Where ψ is the steering angle and $e_\theta = \theta_2 - \theta_1$ is the orientation error. The longitudinal controller outputs the acceleration along the line joining the 2 vehicles, which is subsequently converted to the overall acceleration.
These 2 controllers work parallelly independently, controlling the acceleration and steering angle, respectively, which ensures platooning even during the lane change manoeuvre.

III. Simulation Results

Fig2. shows the velocity-time curve for the lead vehicle and 3 followers when a small positive acceleration is given at 35 seconds. The platoon quickly readjusts its velocities and reaches steady state. Fig3(a) depicts the lane change behavior via y-position vs time plots of a 3-vehicle platoon. The width of the lane is assumed to be 3.2m. The difference of relative distance between vehicles and the safe distance (termed as distance error) during lane change manoeuvre initiated at time =25s is shown by Fig3(b).

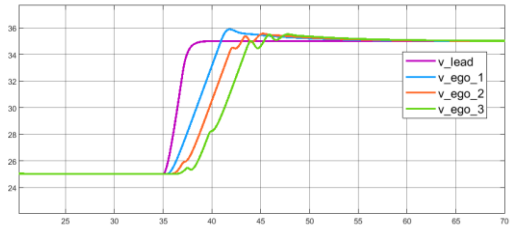


Fig 2. Velocities for a 4-vehicle 1-D platoon

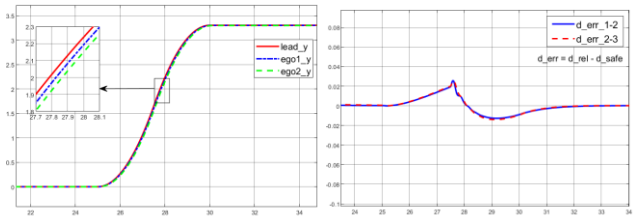


Fig 3. (a) Y-position and (b) Distance error for a 3-vehicle multi-lane platoon

IV. Conclusion

This work presented a novel multi-lane platoon control approach based on the existing ACC infrastructure commonly available in CAVs. Simulations show that the platoon is resilient to disturbances in the lead vehicle's longitudinal velocity and remains intact during lane-changing manoeuvres. Future scope includes the development of planning methods for deciding the feasibility of lane-changing manoeuvres. RL can be explored in this regard. Another important direction is to provide formal guarantees for the stability of platoon.

References

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